

# RayGlides 3W/4W EV Kit Cooling Controller - Manufacturing & Assembly Notes

This document provides production guidelines, board materials, soldering profiles, and wiring assembly specifications.

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## 1. PCB Fabrication Specifications

- \* **Material**: FR-4 TG150 (Glass transition temperature  $\geq 150^{\circ}\text{C}$ )
- \* **Layer Count**: 2 Layers
- \* **Board Thickness**: 1.6 mm
- \* **Copper Weight (Outer)**: 2 oz/ft<sup>2</sup> (70  $\mu\text{m}$ )
- \* **Surface Finish**: HASL Lead-Free (RoHS compliant) or ENIG (Electroless Nickel Immersion Gold)
- \* **Solder Mask Color**: Matte Green or Blue (LPI - Liquid Photoimageable)
- \* **Silkscreen**: White (Lead-Free legibility)
- \* **Minimum Drill Size**: 0.3 mm
- \* **Minimum Trace/Space**: 0.15 mm (6 mils)

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## 2. Component Assembly Instructions

- \* **Solder Paste**: SAC305 (96.5% Sn, 3% Ag, 0.5% Cu) Lead-Free solder paste.
- \* **SMD Reflow Profile**: Conforms to IPC/JEDEC J-STD-020E standard:
- \* **Preheat Temp**:  $150^{\circ}\text{C}$  to  $200^{\circ}\text{C}$  for 60-120 seconds
- \* **Time Above Liquidus** ( $217^{\circ}\text{C}$ ): 60-90 seconds
- \* **Peak Reflow Temp**:  $245^{\circ}\text{C}$  to  $250^{\circ}\text{C}$  for 20-30 seconds
- \* **Manual Assembly**:
- \* Through-hole connectors J1, J2, J3, J4, J5 and temperature sensor U4 can be wave-soldered or hand-soldered at  $350^{\circ}\text{C}$  using lead-free wire (Sn96.5/Ag3.0/Cu0.5).
- \* Ensure the exposed thermal pad of U1 (LM5017) is fully reflowed to the PCB thermal via pad to prevent regulator overheating.

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## 3. Wiring and Harness Specifications

1. **Input Power Harness (J1)**:
  - \* **Connector Shell**: Molex Micro-Fit 3.0 2-circuit housing (43645-0200)
  - \* **Terminal Crimp Pin**: Molex male/female tin crimp terminal (43030-0007)
  - \* **Wire Gauge**: **20 AWG** (Teflon or silicone insulation rated for 300V,  $105^{\circ}\text{C}$  operational temperature)

2. **Fan Power Harness (J2)**:

- \* **Connector Shell**: Molex Micro-Fit 3.0 2-circuit housing (43645-0200)
- \* **Wire Gauge**: **20 AWG** or **22 AWG** (Red: +12V, Black: switched GND return)

3. **Sensor Cable Harness (J3)**:

- \* **Connector Shell**: Molex Micro-Fit 3.0 3-circuit housing (43645-0300)
- \* **Wire Gauge**: **24 AWG** (shielded 3-conductor cable; shield must be connected to GND\_SENS at the PCB side only to minimize EMI loop pickup)

4. **CAN Bus Interface Harness (J4)**:

- \* **Connector Shell**: Molex Micro-Fit 3.0 2-circuit housing (43645-0200)
- \* **Wire Gauge**: **24 AWG** (twisted pair cable, 120 $\Omega$  nominal characteristic impedance)